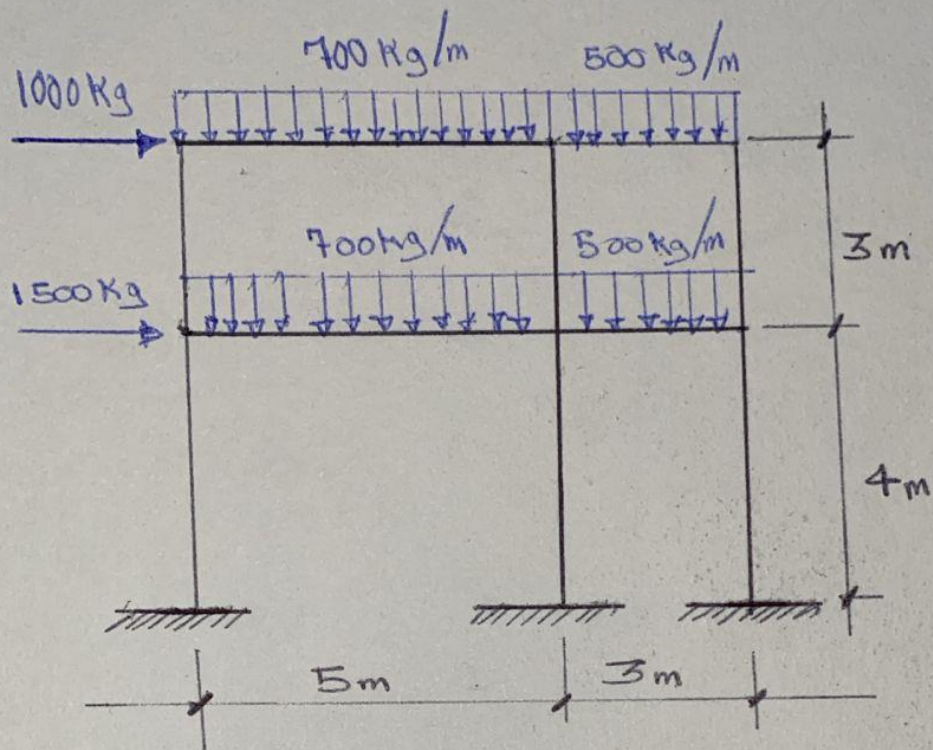




ELEMENTOS

- COLUMNAS 25x25 CM
- VIGAS 25x50 CM
-

MATERIAL

- CONCRETO - 210 Kg/cm²

$\Delta_1 =$

$$\left\{ \begin{array}{l} U_1 z \\ U_1 x \\ \theta_1 \\ \\ U_2 z \\ U_2 x \\ \theta_1 \\ \\ U_3 z \\ U_3 x \\ \theta_3 \\ \\ \dots \end{array} \right\}$$

$$U_1 z = \Delta x_{11}$$

$$U_1 x = \text{CORTE}$$

$$\theta = \text{GIRO}$$

VECTOR DE FUERZAS CONOCIDAS

$$\{F\} = \left\{ \begin{array}{l} 1 \left\{ \begin{array}{l} 1750 \text{ kg} \\ 1000 \text{ kg} \\ 1958.33 \text{ kg/m} \end{array} \right. \\ 2 \left\{ \begin{array}{l} \\ \\ \end{array} \right. \end{array} \right.$$

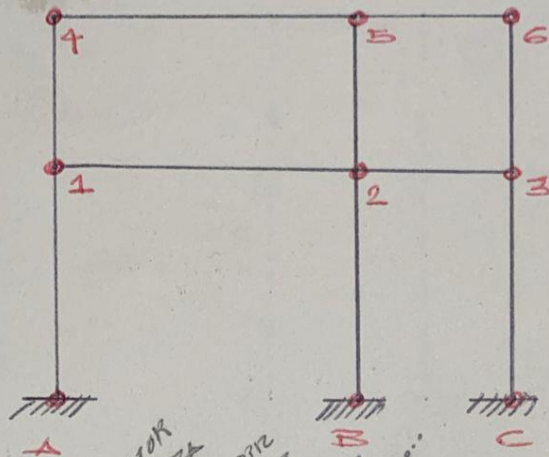
$$F = \frac{WL}{2} = \frac{900 \text{ kg/m} \times 5 \text{ m}}{2} = 1750 \text{ kg}$$

$$M = \frac{WL^2}{12} = \frac{900 \text{ kg/m} \times (5 \text{ m})^2}{12} = 1958.33 \text{ kg/m}$$

$$F = \frac{WL}{2} = \frac{500 \text{ kg/m} \times 3 \text{ m}}{2} = 750 \text{ kg}$$

$$M = \frac{WL^2}{12} = \frac{500 \text{ kg/m} \times (3 \text{ m})^2}{12} = 345 \text{ kg/m}$$

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VECTOR FUERZA
MATRIZ RIGIDEZ
MATRIZ DESPLAZ...

$$\{F\} = \{k\} \{u\}$$

VECTOR DE DESPLAZAMIENTO

$$\{u\} = \begin{Bmatrix} \{\Delta_1\} \\ \{\Delta_2\} \\ \{\Delta_3\} \\ \{\Delta_4\} \\ \{\Delta_5\} \\ \{\Delta_6\} \end{Bmatrix}$$

— MODULO DE ELASTICIDAD ES CONSTANTE

